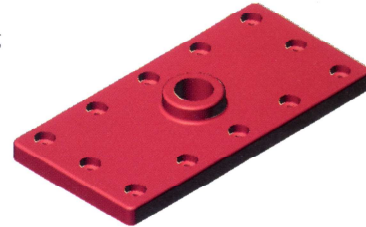


**Exercise 19:
Linear Patterns**

Create feature patterns in this part using a Linear Pattern with Spacing and Instances or Up to reference.

This lab uses the following skills:

- *Linear Pattern* on page 173
- *Skipping Instances* on page 175
- *Up To Reference* on page 190

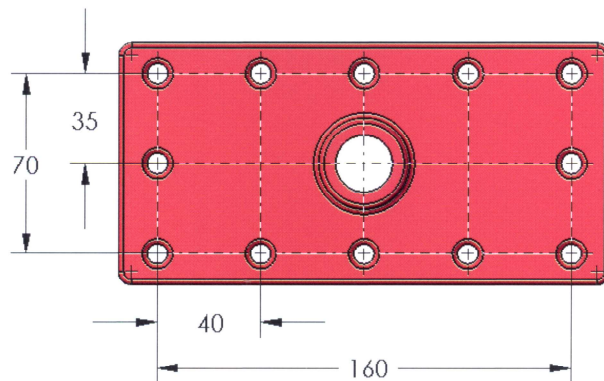
**Procedure**

Open an existing part.

- 1 Open the part *Linear*.**
The part includes the “seed” feature used in the patterns.



- 2 Linear pattern.**
Create a pattern using the seed feature. Use the dimensions below.



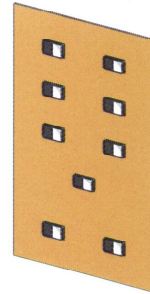
- 3 Save and close the part.**

Exercise 20: Sketch Driven Patterns

Create feature patterns using a Sketch Driven Pattern. The model is an elevator panel.

This lab uses the following skills:

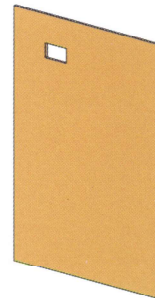
- *Sketch Driven Patterns* on page 192



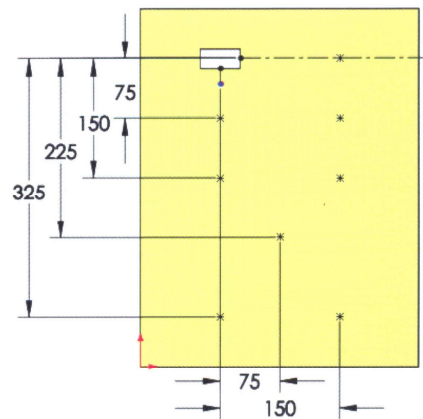
Procedure

Open an existing part.

- 1 **Open the part *Sketch Driven Pattern*.**
The part includes the “seed” feature used in the pattern.



- 2 **Sketch driven pattern.**
Use the dimensions shown to define the sketch. Use the sketch to create a sketch driven pattern.



Thanks to Marcus Brown of MLC CAD Systems for submitting this example.

- 3 **Save and close the part.**

Exercise 21: Skipping Instances

Complete this part using the information and dimensions provided.

This lab reinforces the following skills:

- *Linear Pattern* on page 173
- *Skipping Instances* on page 175
- *Mirror Patterns* on page 186

Units: **millimeters**

Procedure

Create a new part.

1 New part.

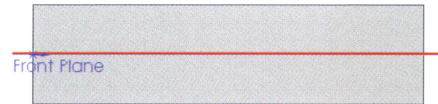
Create a new mm part

2 Base feature.

Create a block

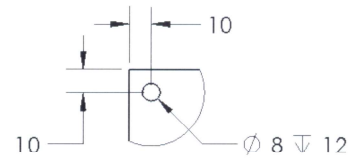
75mmx380mmx20mm.

It will be useful to have a plane centered along the long direction.



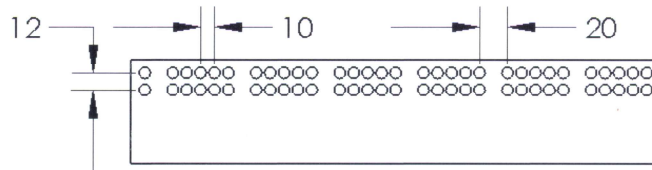
3 Seed.

Create the seed feature using the Hole Wizard and an ANSI MM drill.



4 Pattern.

Pattern the hole, skipping instances as shown in the diagram below.



5 Pattern of a pattern.

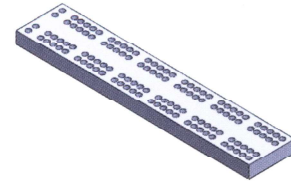
Mirror the pattern to create a symmetrical arrangement of holes.



6 Change.

Change the hole to **4mm** diameter and rebuild.

7 Save and close the part.

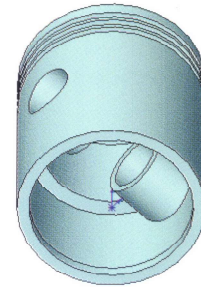


Exercise 22: Linear and Mirror Patterns

Complete this part using the information and dimensions provided.

This lab reinforces the following skills:

- *Linear Pattern* on page 173
- *Mirror Patterns* on page 186
- *Patterning a Solid Body* on page 187



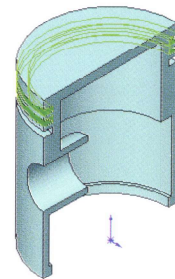
Procedure

Open an existing part.

1 Open the part *Linear & Mirror*.

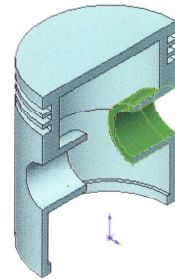
2 Linear pattern.

Using the existing feature, create a **Linear Pattern** that results in three grooves that are spaced **0.20"**.



3 Mirror features.

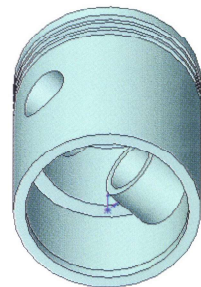
Using a single pattern feature, create the duplicate boss and cut as shown.



4 Symmetry.

Use a third pattern feature to create the full model from the half model using **Bodies to Mirror**.

5 Save and close the part.



**Exercise 23:
Circular
Patterns**

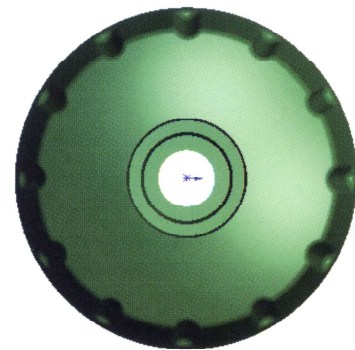
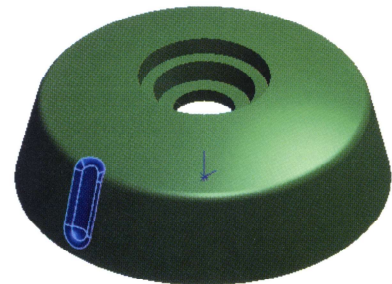
Complete this part using the information and dimensions provided.

This lab reinforces the following skills:

- *Circular Patterns* on page 179

Procedure

Open the existing part *Circular*.
Use an equally spaced circular pattern to pattern the cut and fillet for 12 total instances.

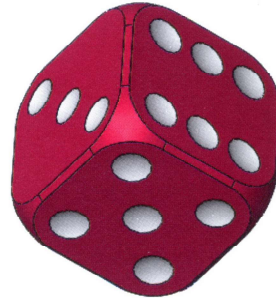


Exercise 24: Axes and Multiple Patterns

Complete this part using the information and dimensions provided.

This lab reinforces the following skills:

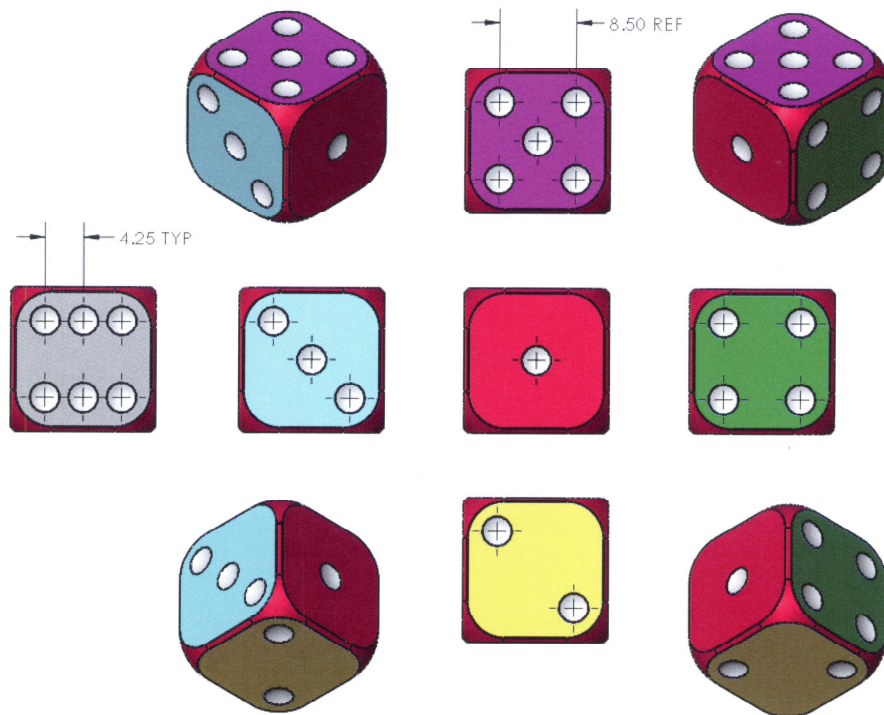
- *Axes* on page 180
- *Linear Pattern* on page 173
- *Circular Patterns* on page 179
- *Sketch Driven Patterns* on page 192



Procedure

Open the existing part Single Die. Use the drawing below to pattern the Dot feature on the sides as shown.

The face colors are used to help distinguish between the individual faces. See step 4 on page 206 for *removing* the colors.



Note

Orient the part as shown in each step before starting work.